Задание 3.2

```
Запуск программы:
  загрузка исходного окружения

--> Xn=-3.5;dX=1.5;Xk=4.5;

--> X=Xn:dX:Xk
 X  =

  -3.5  -2.  -0.5   1.   2.5   4.


--> Y=sin(X/2)
 Y  =


         column 1 to 3

  -0.9839859  -0.841471  -0.247404

         column 4 to 6

   0.4794255   0.9489846   0.9092974


--> A=0:5
 A  =

   0.   1.   2.   3.   4.   5.


--> 0:5
 ans  =

   0.   1.   2.   3.   4.   5.


--> ans/2+%pi
 ans  =
```

```
   3.1415927   3.6415927   4.1415927

         column 4 to 6

   4.6415927   5.1415927   5.6415927


--> V=[1 2 3 4 5]
 V  =

   1.   2.   3.   4.   5.


--> W=[1.1,2.3,-0.1,5.88]
 W  =

   1.1   2.3  -0.1   5.88


--> A=[1, 1, 1, 1, 1,]
A=[1, 1, 1, 1, 1,]
                 ^^
Ошибка: syntax error, unexpected ]

--> A=[1 1 1 1 1 ]
 A  =

   1.   1.   1.   1.   1.


--> X=[1;2;3]
 X  =

   1.
   2.
   3.


--> X=[1;2;3;4;5;6;7;8;9;10]
 X  =

   1.
   2
```

```
--> X=[1;2;3;4;5;6;7;8;9;10]
 X  =

   1.
   2.
   3.
   4.
   5.
   6.
   7.
   8.
   9.
   10.


--> W=[1.1,2.3,-0.1,5.88];

--> W=[1.1,2.3,-0.1,5.88];

--> W(1)+2*W(3)
 ans  =

   0.9


--> A=[1 2 3;4 5 6;7 8 9]
 A  =

   1.   2.   3.
   4.   5.   6.
   7.   8.   9.


--> B=[1 2 3 4;5 6 7 8;9 10 11 12]
 B  =

   1.   2.    3.    4.
   5.   6.    7.    8.
   9.   10.   11.   12.
```

```
--> V=[v1; v2; v3]
 V  =

   1.   2.   3.
   4.   5.   6.
   7.   8.   9.


--> M=[V V V]
 M  =

   1.   2.   3.   1.   2.   3.   1.   2.   3.
   4.   5.   6.   4.   5.   6.   4.   5.   6.
   7.   8.   9.   7.   8.   9.   7.   8.   9.


--> M=[V;V]
 M  =

   1.   2.   3.
   4.   5.   6.
   7.   8.   9.
   1.   2.   3.
   4.   5.   6.
   7.   8.   9.


--> M=[V;V]
 M  =

   1.   2.   3.
   4.   5.   6.
   7.   8.   9.
   1.   2.   3.
   4.   5.   6.
   7.   8.   9.
```

```
--> A=[5 7 6 5; 7 10 8 7;6 8 10 9;5 7 9 10]
 A  =

   5.   7.    6.    5.
   7.   10.   8.    7.
   6.   8.    10.   9.
   5.   7.    9.    10.


--> A(:,2)
 ans  =

   7.
   10.
   8.
   7.


--> A(3,:)
 ans  =

   6.   8.   10.   9.


--> M=A(3:4,2:3)
 M  =

   8.   10.
   7.   9.


--> A(:,2)=[]
 A  =

   5.   6.    5.
   7.   8.    7.
   6.   10.   9.
   5.   9.    10.


--> A(3,:)=[]
 A  =

   5.   6.   5.
```

```
--> A(3,:)=[]
 A  =

   5.   6.   5.
   7.   8.   7.
   5.   9.   10.


--> v=M(:)
 v  =

   8.
   7.
   10.
   9.


--> b=v(2:4)
 b  =

   7.
   10.
   9.


--> b(2)=[];

--> A=[1 2 0;-1 3 1;4 -2 5];

--> B=[-1 0 1;2 1 1;3 -1 -1];

--> (A+B)^2-2*A*(1/2*B'-A)

Неопределённая переменная: B'

--> (A'+B)^2-2*A*(1/2*B'-A)
 ans  =

   10.   8.    24.
   11.   20.   35.
   63.  -30.   68.
```

```
--> A=[3 2;4 3];

--> B=[-1 7;3 5];

--> X=A\B
 X  =

  -9.    11.
   13.  -13.


--> A*X-B
 ans  =

   0.   0.
   0.   0.


--> X=B/A
 X  =

  -31.   23.
  -11.   9.


--> x=[0.1 -2.2 3.14 0 -1];

--> sin(x)
 ans  =


         column 1 to 4

   0.0998334  -0.8084964   0.0015927   0.

         column 5

  -0.841471


--> D=[1 2;3 4;5 6];

--> matrix(D,2,3)
 ans  =
```

```
--> D=[1 2;3 4;5 6];

--> matrix(D,2,3)
 ans  =

   1.   5.   4.
   3.   2.   6.


--> matrix(D,3,2)
 ans  =

   1.   2.
   3.   4.
   5.   6.


--> matrix(D,1,6)
 ans  =

   1.   3.   5.   2.   4.   6.


--> matrix(D,6,1)
 ans  =

   1.
   3.
   5.
   2.
   4.
   6.


--> ones(1,3)
 ans  =

   1.   1.   1.


--> ones(2,2)
 ans  =

   1.   1.
```

```
--> m=3; n=2;

--> X=ones(m,n)
 X  =

   1.   1.
   1.   1.
   1.   1.


--> M=[1 2 3;4 5 6]
 M  =

   1.   2.   3.
   4.   5.   6.


--> Y=ones(M)
 Y  =

   1.   1.   1.
   1.   1.   1.


--> zeros(3,2)
 ans  =

   0.   0.
   0.   0.
   0.   0.


--> M=[1 2 3 4 5];

--> Z=zeros(M)
 Z  =

   0.   0.   0.   0.   0.


--> eye(3,3)
 ans  =

   1.   0.   0.
```

```
--> eye(3,3)
 ans  =

   1.   0.   0.
   0.   1.   0.
   0.   0.   1.


--> eye(5,1)
 ans  =

   1.
   0.
   0.
   0.
   0.


--> m=3; n=4;

--> E=eye(m,n)
 E  =

   1.   0.   0.   0.
   0.   1.   0.   0.
   0.   0.   1.   0.


--> M=[0 1;2 3];

--> E=eye(M)
 E  =

   1.   0.
   0.   1.


--> M=[1 2;3 4;5 6]; E=eye();

--> A=E+M
 A  =

   2.   2.
   3.   5.
```

```
--> M-E
 ans  =

   0.   2.
   3.   3.
   5.   6.


--> rand(2,2)
 ans  =

   0.2113249   0.0002211
   0.7560439   0.3303271


--> R=rand(2,2,2)
 R  =

(:,:,1)

   0.6653811   0.8497452
   0.6283918   0.685731
(:,:,2)

   0.8782165   0.5608486
   0.068374    0.6623569


--> rand()
 ans  =

   0.7263507


--> rand()
 ans  =

   0.1985144


-->

--> rand()
```

```
--> A=sparse([1 3;3 2;3 5],[4,5,6])
 A  =

(  3,  5) sparse matrix

(  1,  3)         4.
(  3,  2)         5.
(  3,  5)         6.


--> full(A)
 ans  =

   0.   0.   4.   0.   0.
   0.   0.   0.   0.   0.
   0.   5.   0.   0.   6.


--> -M=hypermat([2 2 3],0:11)
-M=hypermat([2 2 3],0:11)
  ^^
Ошибка: syntax error, unexpected =, expecting en

--> M=hypermat([2 2 3],0:11)
ВНИМАНИЕ: Функция hypermat устарела.
ВНИМАНИЕ: Используйте matrix.
ВНИМАНИЕ: Эта функция будет окончательно удалена
 M  =

(:,:,1)

   0.   2.
   1.   3.
(:,:,2)

   4.   6.
   5.   7.
(:,:,3)

   8.   10.
   9.   11.
```

```
--> diag(V)
 ans  =

   1.   0.   0.
   0.   2.   0.
   0.   0.   3.


--> diag(V,1)
 ans  =

   0.   1.   0.   0.
   0.   0.   2.   0.
   0.   0.   0.   3.
   0.   0.   0.   0.


--> diag(V,-1)
 ans  =

   0.   0.   0.   0.
   1.   0.   0.   0.
   0.   2.   0.   0.
   0.   0.   3.   0.


--> A=[-1 2 0 ;2 1 -1 ;2 1 3]
 A  =

  -1.   2.   0.
   2.   1.  -1.
   2.   1.   3.


--> diag(A)
 ans  =

  -1.
   1.
   3.


--> A=[1 2;3 4]; B=[5 6 ;7 8];
```

```
--> cat(2,A,B)
 ans  =

   1.   2.   5.   6.
   3.   4.   7.   8.


--> cat(1,A,B)
 ans  =

   1.   2.
   3.   4.
   5.   6.
   7.   8.


--> A=[1 2 3;4 5 6;7 8 9]
 A  =

   1.   2.   3.
   4.   5.   6.
   7.   8.   9.


--> tril(A)
 ans  =

   1.   0.   0.
   4.   5.   0.
   7.   8.   9.


--> tril(A,0)
 ans  =

   1.   0.   0.
   4.   5.   0.
   7.   8.   9.


--> tril(A,-2)
 ans  =
```

```
--> cat(2,A,B)
 ans  =

   1.   2.   5.   6.
   3.   4.   7.   8.


--> cat(1,A,B)
 ans  =

   1.   2.
   3.   4.
   5.   6.
   7.   8.


--> A=[1 2 3;4 5 6;7 8 9]
 A  =

   1.   2.   3.
   4.   5.   6.
   7.   8.   9.


--> tril(A)
 ans  =

   1.   0.   0.
   4.   5.   0.
   7.   8.   9.


--> tril(A,0)
 ans  =

   1.   0.   0.
   4.   5.   0.
   7.   8.   9.


--> tril(A,-2)
 ans  =
```

```
 ans  =

   2.


--> V=[-1 0 3 -2 1 -1 1];

--> length(V)
 ans  =

   7.


--> [1 2 3;4 5 6];

--> length(ans)
 ans  =

   6.


--> M=[1 2 3;4 5 6;7 8 9];

--> Y=sum(M)
 Y  =

   45.


--> S1=sum(M,1)
 S1  =

   12.   15.   18.


--> S2=sum(M,2)
 S2  =

   6.
   15.
   24.
```

```
--> V=[-1 0 3 -2 1 -1 1];

--> sum(V)
 ans  =

   1.


--> a=[1 2 3];b=[2 0 1];

--> sum(a.*b)
 ans  =

   5.


--> prod(M)
 ans  =

   362880.


--> p1=prod(M,1)
 p1  =

   28.   80.   162.


--> p2=prod(M,2)
 p2  =

   6.
   120.
   504.


--> V=[1,2,3];

--> prod(V)
 ans  =

   6.
```

```
--> V=[-1 0 3 -2 1 -1 1];

--> sum(V)
 ans  =

   1.


--> a=[1 2 3];b=[2 0 1];

--> sum(a.*b)
 ans  =

   5.


--> prod(M)
 ans  =

   362880.


--> p1=prod(M,1)
 p1  =

   28.   80.   162.


--> p2=prod(M,2)
 p2  =

   6.
   120.
   504.


--> V=[1,2,3];

--> prod(V)
 ans  =

   6.
```

```
--> prod(V)
 ans  =

   6.


--> M=[5 0 3;2 7 1;0 4 9];

--> max(M)
 ans  =

   9.


--> max(M,'r')

Неопределённая переменная: 'r'

--> -->max(M,'r)
-->max(M,'r)
  ^^
Ошибка: syntax error, unexpected >

--> max(M',r')

Неопределённая переменная: r

--> -->A=[5 10 3 2 7 1 25 4 0];
-->A=[5 10 3 2 7 1 25 4 0];
  ^^
Ошибка: syntax error, unexpected >

--> A=[5 10 3 2 7 1 25 4 0];

--> ax(M,'r')

Неопределённая переменная: ax

--> max(M,'r')
 ans  =

   5.   7.   9.
```

```
--> max(M,'r')
 ans  =

   5.   7.   9.


--> max(M)
 ans  =

   9.


--> M=[1 0 2;3 2 1;0 3 1];

--> rank(M)
 ans  =

   3.


--> Z=[1 2 2;0 1 3;2 4 4];

--> rank(Z)
 ans  =

   2.
```